

Clarifying the metabolic calculation reference

Lean · Product requirements

Draft · September 2026 · Independent work

1. Summary

Users who select "Other" as their gender during Lean's onboarding are assigned the male metabolic formula without being told. For a user with female physiology this means a daily target roughly 250 kcal above their actual requirement, which undermines the core promise of the product.

This document states the problem, frames two candidate interventions as a comparative experiment, sets out requirements and acceptance criteria, and covers release checks and prioritisation.

2. Problem

Evidence

Onboarding was completed three times from a clean install. Gender was the only variable changed; height, weight, age, target weight, weekly loss rate, workout frequency and diet type were held constant.

Gender	Calories	Protein	Carbs	Fat
Female	1758	154 g	154 g	59 g
Male	2008	176 g	176 g	67 g
Other	2008	176 g	176 g	67 g

The "Other" result is identical to "Male" — a difference of 250 kcal per day, roughly 14%, against the female result.

Why this matters

- Lean's published "Sources of Recommendations" page lists only two Harris-Benedict variants, female and male. Nothing states which coefficients apply to the third option.
- The user cannot see which formula was applied, and cannot change it.
- Someone following the plan exactly will not reach the promised result. This is the kind of gap that produces refund requests and low ratings.

Affected users

Users who select "Other" during onboarding. The size of this segment cannot be measured from outside and needs to be pulled from existing analytics before the work is scheduled. Even if the segment is small, the per-user impact is high.

3. Goals and scope

Goals:

- Users who select "Other" continue with a metabolic reference appropriate to them.
- The applied reference is visible to the user and can be changed.

Out of scope:

- Replacing Harris-Benedict with a different BMR formula.

- Changing the number or naming of gender options.
- Documenting macro ratios per diet type (tracked separately).

4. User story

As a user who selects "Other" during onboarding, I want to know which metabolic reference my calorie and macro targets are based on — and to change it if needed — because a plan built on the wrong reference will not get me to my goal.

5. Solution and experiment design

Two interventions are possible, and the trade-off between accuracy and friction cannot be settled in advance. Both are therefore tested in a single experiment.

Arm	Behaviour	Rationale
Control	Current behaviour: the default is applied silently.	Reference arm.
Arm A	Follow-up question: "Which reference should we use for your calculation?" Two options plus a "Skip for now" link.	Highest accuracy. Cost: one more step in the flow.
Arm B	No question; the applied default is stated on the results screen with a link to settings.	No friction. Cost: the problem persists if the user never opens settings.

In both arms the additional screen is shown only to users who selected "Other"; no other user's flow changes.

The question asks about the calculation reference rather than gender identity, and explains in one line why it is being asked. Answering is optional; if skipped, the default is applied and stated on screen.

6. Requirements and acceptance criteria

R-1 — Reference question screen (arm A)

Given the user selected "Other" as gender during onboarding and is in arm A,

When they move past the gender step,

Then a screen asks which reference to use, showing two reference options, a one-line explanation of why it is asked, and a "Skip for now" link.

R-2 — Selection applied to the calculation

Given the user has selected a reference,

When the plan is calculated,

Then calorie and macro targets are produced using the coefficients of the selected reference, and the selection is saved to the user profile.

R-3 — Skip behaviour

Given the user tapped "Skip for now",

When the plan screen opens,

Then the plan is calculated with the default reference and the screen states which reference was used.

R-4 — Reference stated on results (arm B)

Given the user selected "Other" and is in arm B,

When the plan results screen opens,

Then a line states which reference was used, with a link to settings.

R-5 — Changing the reference in settings (both arms)

Given the user changes the calculation reference in settings,

When they save the change,

Then daily calorie and macro targets are recalculated, meal logs recorded before the change are preserved, and the user is told their targets have been updated.

R-6 — Analytics

Given the user sees the reference question, answers it, skips it, or changes it in settings,

When the interaction occurs,

Then a distinct analytics event is recorded for each, tagged with the experiment arm.

7. Experiment plan

Hypothesis

Users who can either choose or at least see the metabolic reference applied to them will continue with a correctly calculated target, which over time should improve retention and reduce refunds. The effect of the extra step on onboarding completion is expected to be negligible.

Audience

New users who select "Other" during onboarding. Existing users are out of scope; for them, the settings option (R-5) is sufficient.

Metrics

Type	Metric	Note
Primary	Share of users continuing with a correct reference	In arm A, users who answered the question; in arm B, users who changed it in settings; zero by definition in control. This metric decides the outcome.
Guardrail	Onboarding completion rate	A significant drop in arm A means the extra step carries a real cost.
Guardrail	Paywall view → subscribe conversion	Confirms the flow change does not touch revenue.
Secondary	D7 and D30 retention	Not a decision metric given the segment size; watched as a directional signal.
Secondary	Refund rate	A lagging signal, monitored after the experiment ends.

Sample size and duration

The primary metric is a fast-accumulating rate, so it needs far less volume than retention would. The "Other" segment is small, though, so duration has to be calculated up front:

- Pull the daily count of users selecting "Other" over the last 30 days from existing analytics.
- Set the minimum detectable effect. Since the primary metric is zero by definition in control, arms A and B are compared against each other.
- Run a separate power calculation for the guardrails. The minimum drop worth detecting in onboarding completion is usually smaller than the effect on the primary metric, so the guardrail is what sets the duration.

Note: the traffic data needed for this calculation is not available externally. This section should be filled in with real figures before the work is scheduled.

Decision rule

- If arm A or B beats control on the primary metric with no significant drop on any guardrail, the winning arm ships.
- If arm A wins on the primary metric but onboarding completion drops significantly, arm B is preferred.
- If neither arm differs from control on the primary metric, nothing ships and the source of the problem is reassessed.

8. Release readiness

Regression checks

- Complete onboarding end to end with each of the three gender options; verify the calorie and macro output against the expected coefficients.
- Test the skip path; verify the default is applied and stated on screen.
- Verify that changing the reference in settings does not corrupt existing meal logs.
- Go back to the gender step and change the selection; verify the question screen appears or disappears correctly.
- Close and reopen the app mid-onboarding; verify it resumes at the right step.
- Verify the paywall and subscription flow are unaffected.
- Verify the new copy exists in every supported language.

Post-release monitoring

- First 48 hours: crash and error rate on the new screen, and consistency between impressions and responses.
- First week: stop the experiment if onboarding completion or paywall conversion falls below guardrail thresholds.
- Ongoing: primary metric and retention by segment.

9. Prioritisation

In favour:

- High per-user impact: a miscalculated target undermines the core promise of the product.

- Low build cost: the formulas and personalisation plumbing already exist; no new calculation is being written.
- Closes a gap between the published methodology and actual product behaviour.

Against:

- The affected segment is probably small, so total revenue impact may be limited and the experiment may take a while to reach significance.

Assessment: schedule this after the segment size is measured. If the segment is larger than expected, it moves up; if small, it stands as a low-cost correctness and trust improvement. Either way, R-5 can ship independently of the experiment and delivers value on its own.

10. Open questions

- How large is the "Other" segment?
- Is the current default a deliberate choice, or the library's default behaviour?
- Does the question copy need legal or people-team review?
- Should existing users who already selected "Other" be notified retroactively?

This document follows from an independent review of the publicly released app. Without access to internal data, assumptions about segment size, sample calculations and existing product decisions are marked as such throughout.